MA2102 2026-2027 Test 2

Time: 50 minutes Total marks: 30

1. Solve

$$x\frac{\partial u}{\partial x} + 2u = 6xy, \quad x > 0,$$

subject to $u(1, y) = y^2$

[6 marks]

2. The function

$$f(x) = x^2, \quad -\pi < x < \pi,$$

is extended periocically with period 2π.

(a) State wheter f is odd or even. Hence, identify which Fourier coefficients vanish.

(b) Calculate the remaining Fourier coefficients and obtain the Fourier series of f.

(c) By evaluating the series at a suitable value of x, deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} - \frac{\pi^2}{6}.$$

(d) State the value to which the Fourier series converges at $x = \pi$.

[2 + 7 + 2 = 12 marks]

3. (a) Find

$$\mathcal{L}^{-1}\left\{\frac{s+5}{s^2+4s+13}\right\}.$$

(b) Use Laplace transforms to solve the initial value problem

$$y'' + 3y' + 2y = 4, \quad y(0) = 0, \quad y'(0) = 0$$

(c) Determining the limiting value of the solution in part (b) as $t \to \infty$.

[4 + 7 + 1 = 12 marks]

END OF QUESTION PAPER